

Large Language Models and Knowledge Representation

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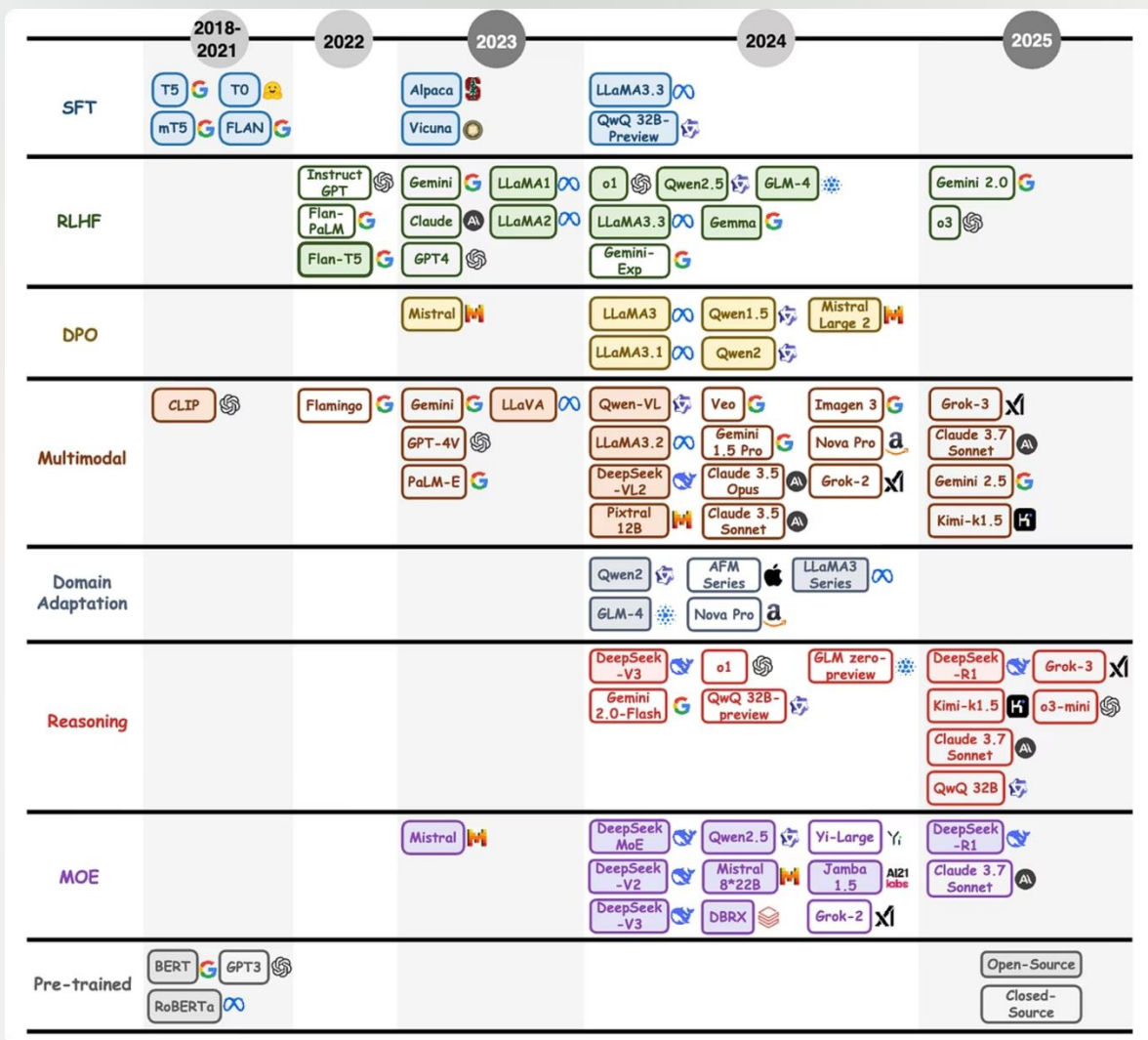
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National Research Council - ISTC**





A Timeline of Large Language Models (LLMs)

The evolution of language models has accelerated dramatically, transforming from simple statistical models to sophisticated AI systems capable of human-like text generation and reasoning.

Tie et al. A Survey on Post-training of Large Language Models - arXiv

Key Questions

How do LLMs work?

Understanding the foundational mechanisms that enable language models to process and generate human-like text

How can LLMs help us?

Practical applications and ways these models can enhance our research and daily activities

LLMs for ontologies?

Exploring how we can leverage LLMs for creating, populating, and managing knowledge representation systems



Outline

01

How LLMs Work

Presented by Misael

- Foundation of Language Models
- Inside a Large Language Model
- Aligning LLMs
- Controlling LLMs

02

LLMs for Knowledge Representation

Presented by Luana

- Prompt engineering
- Ontology generation
- Entity extraction and alignment
- Relation extraction



Distributional Hypothesis

**"The meaning of a word is its use in
in the language"**

Ludwig Wittgenstein

This foundational principle underlies modern natural language processing and explains how machines can learn meaning from patterns in text usage.

What does recent English borrowing "**ongchoi**" mean? mean?

Suppose you see these sentences:

- Ongchoi is delicious **sautéed with garlic**
- Ongchoi is superb **over rice**
- Ongchoi **leaves** with salty sauces

And you've also seen these:

- ...spinach **sautéed with garlic over rice**
- Chard stems and **leaves** are **delicious**
- Collard greens and other **salty** leafy greens



Conclusion:

Ongchoi is a leafy green like spinach, chard, or collard greens

We conclude this based on shared context words like "leaves," "delicious," and "sautéed"

Ongchoi: *Ipomoea aquatica* "Water Spinach"



Yamaguchi, Wikimedia Commons, public domain

Known by many names:

1

空心菜

Chinese name

2

kangkong

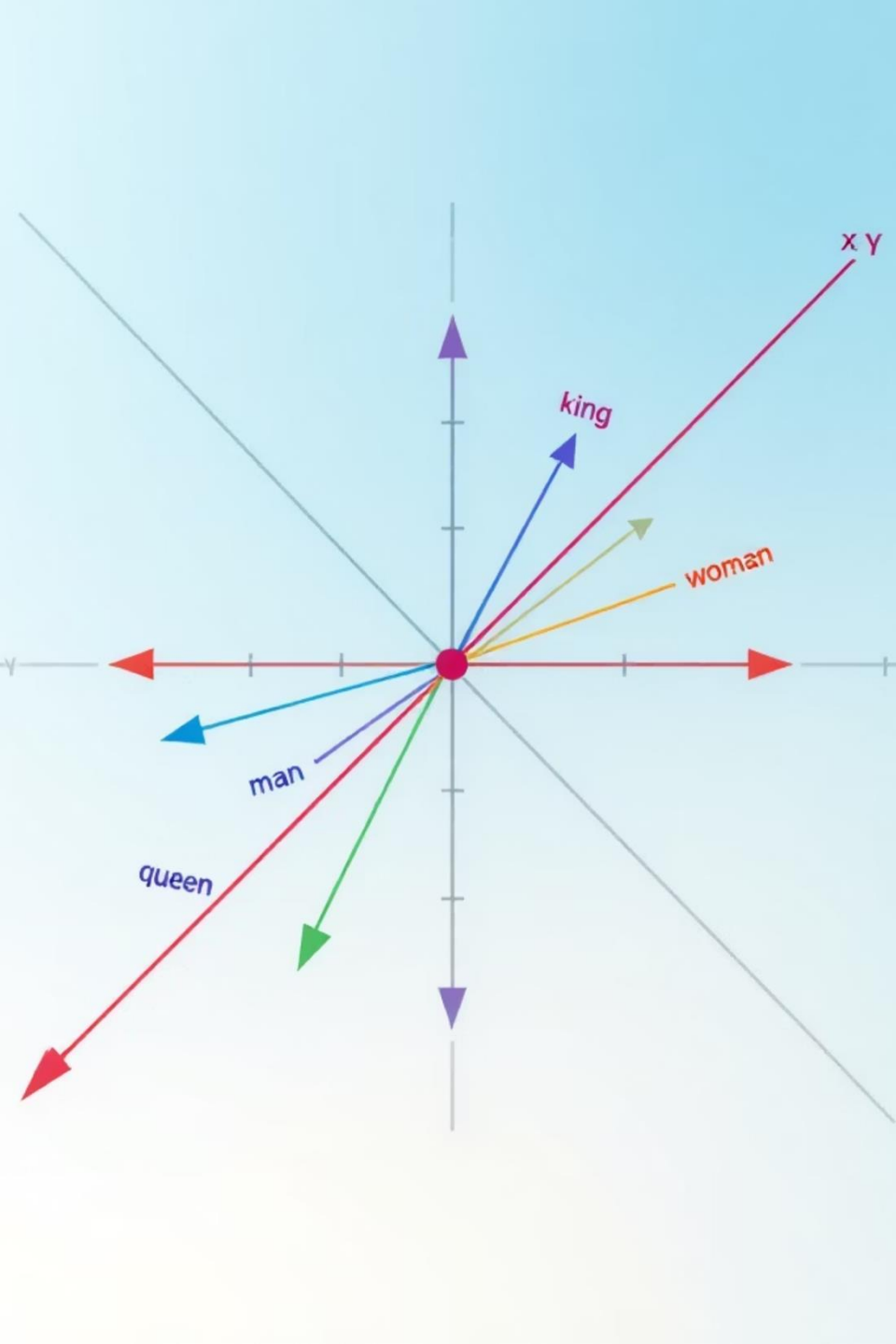
Filipino name

3

rau muống

Vietnamese name

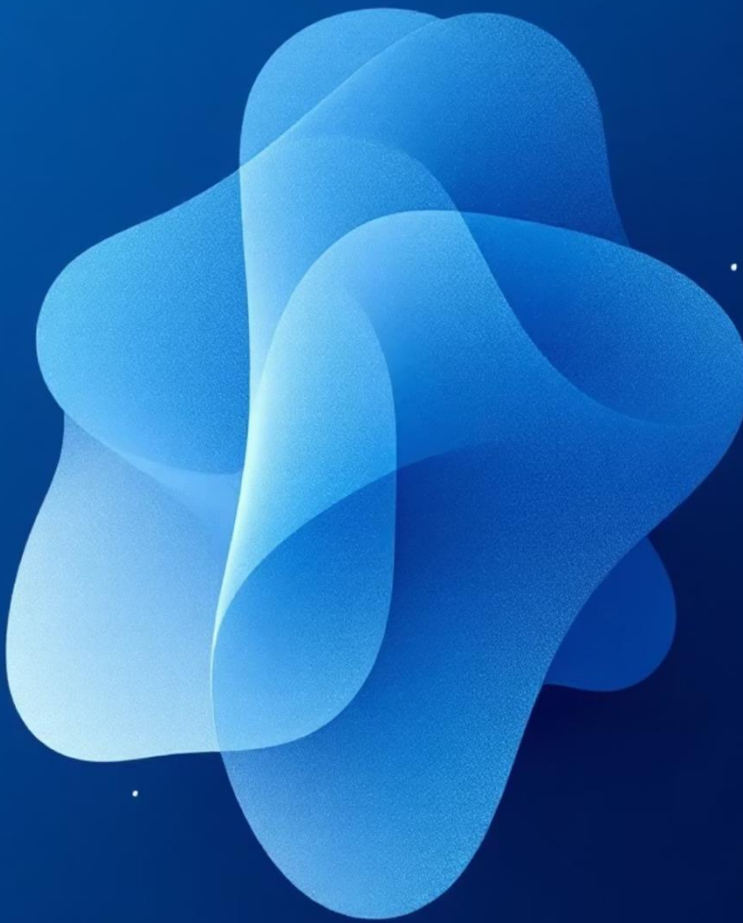
This shows how context clues help us understand meaning



Meaning as a Vector in a Space Space

Modern NLP represents word meanings as mathematical vectors in high-dimensional space, where semantic similarity corresponds to geometric proximity.

Word embeddings typically exist in 300-4096 dimensional spaces, capturing capturing nuanced semantic relationships impossible to visualize directly.

An abstract graphic on the left side of the slide, consisting of several overlapping, rounded, blue shapes that resemble a stylized flower or a cluster of petals. The shapes are in various shades of blue, from light to dark, and are set against a dark blue background.

We Represent the Meaning of a Word Word as a Vector



Embeddings

Called an "embedding" because meaning is embedded into a multidimensional mathematical space



Standard Representation

The foundational way to represent meaning in modern natural language processing systems



Universal Usage

Every modern NLP algorithm uses embeddings as the core representation of word meaning



Capture Word Similarity

Provides a fine grained models of meaning that enable precise semantic similarity similarity calculations

To recap

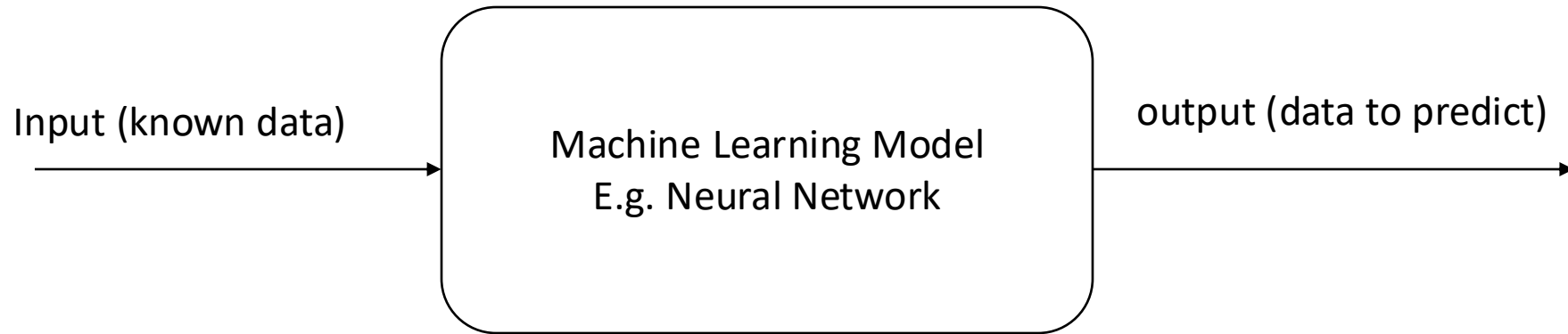
Defining Meaning by Linguistic Distribution

Words gain their meaning from the contexts in which they appear. Analyzing patterns of usage allows models to infer semantic relationships and understand nuances

Meaning as a Multidimensional Vector

This contextual understanding is mathematically encoded into high-dimensional vectors, where proximity in space reflects semantic similarity between words

Machine learning basics

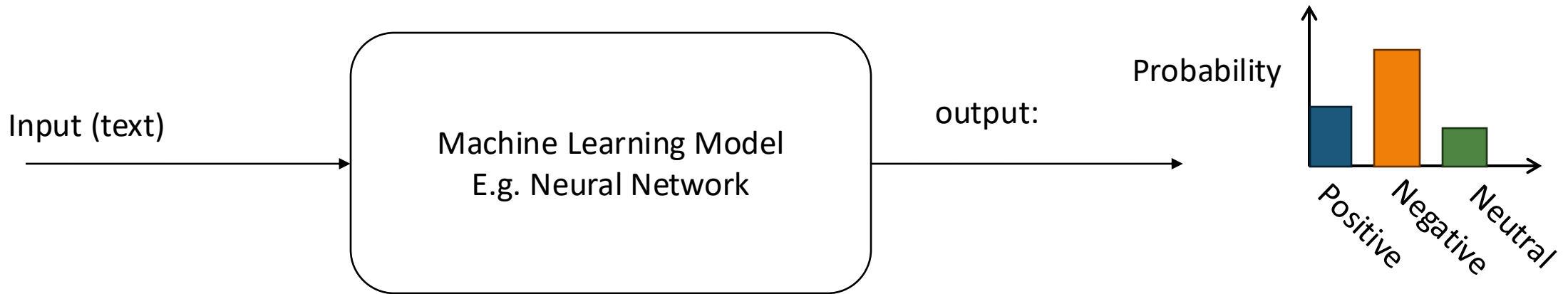


Training: we provide a number of examples with known output. The model learns the relationship between input and output.

In Neural Networks the relationship between the input and the output is given by a function that depend on (a lot of) **parameters**

Inference: we provide the input and the model returns the output

Classification model



Probability distribution: for each possible class, the model provides a probability estimate that it is the correct class

- Each probability is a value between 0 and 1
- The sum of the probabilities assigned to all classes is 1

Probabilistic Language Modeling

Goal: compute the probability of an upcoming word given previous words:

$$P(w_n | w_1, w_2, \dots, w_{n-1})$$

A model that computes $P(w_n | w_1, w_2 \dots w_{n-1})$ is called a **Language Model**

Example

Given the text “Tart with apricot...”, we want to predict the next word (token):

$P(\text{“jam”} \mid \text{“Tart”, “with”, “apricot”})$

we want it to be high

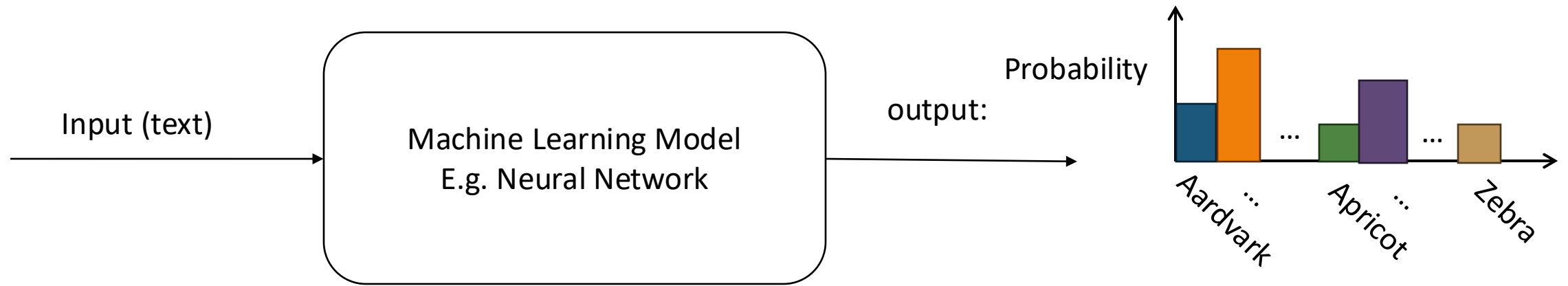
$P(\text{“sausage”} \mid \text{“Tart”, “with”, “apricot”})$

we want it to be very low

$P(\text{“satellite”} \mid \text{“Tart”, “with”, “apricot”})$

we want it to be very low

Implementing a Language model



Given an input text, the model estimates the probability distribution of the next word over the entire vocabulary

Training

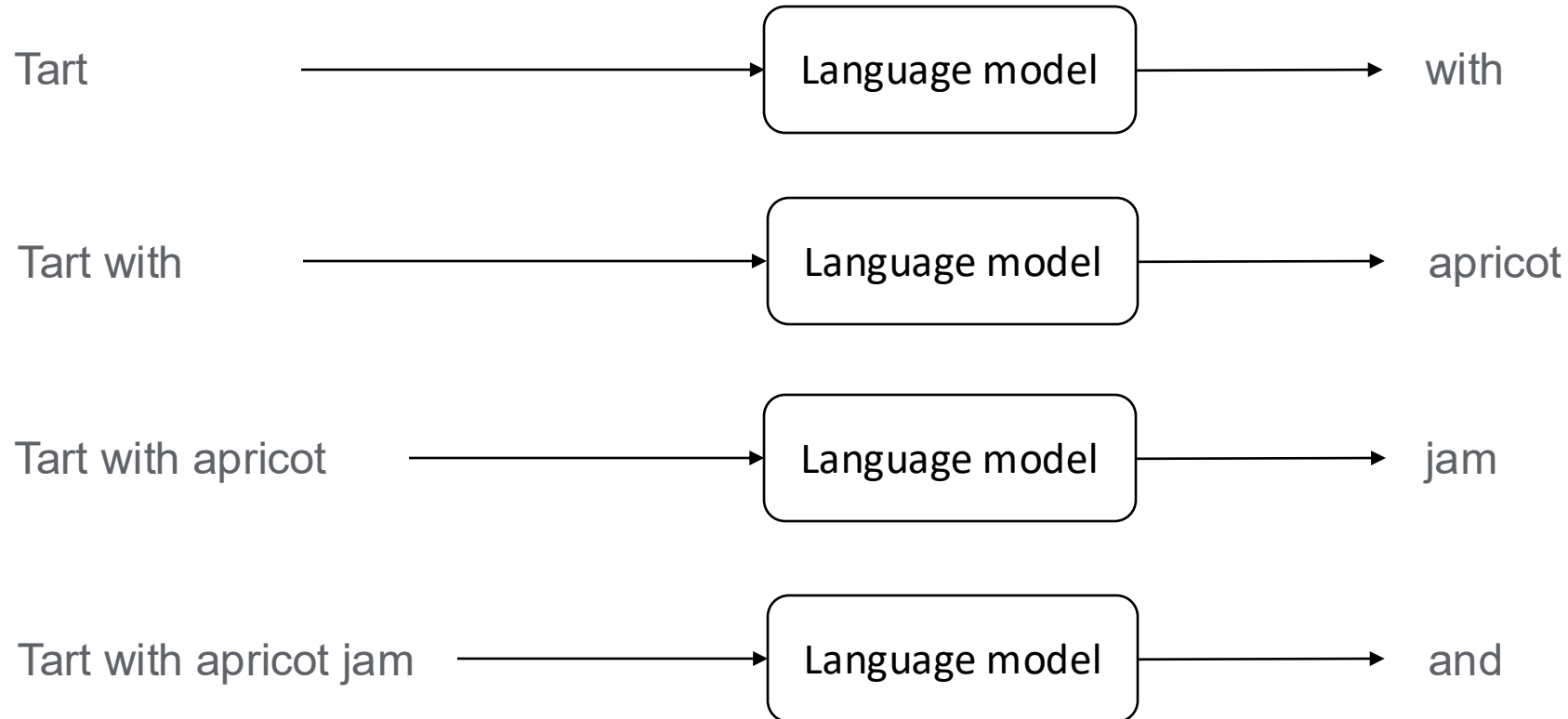
Given a large text corpus, we can build many samples by considering one word at a time from the first to the last word of the corpus.

For each of those words, we consider the text preceeding the word as the **input** and the word itself as the **output**

Training

Given a long text we can split it in many windows, each with a partial text as *input* and the next word as *output*

E.g. "Tart with apricot jam and..."



Language model for text generation

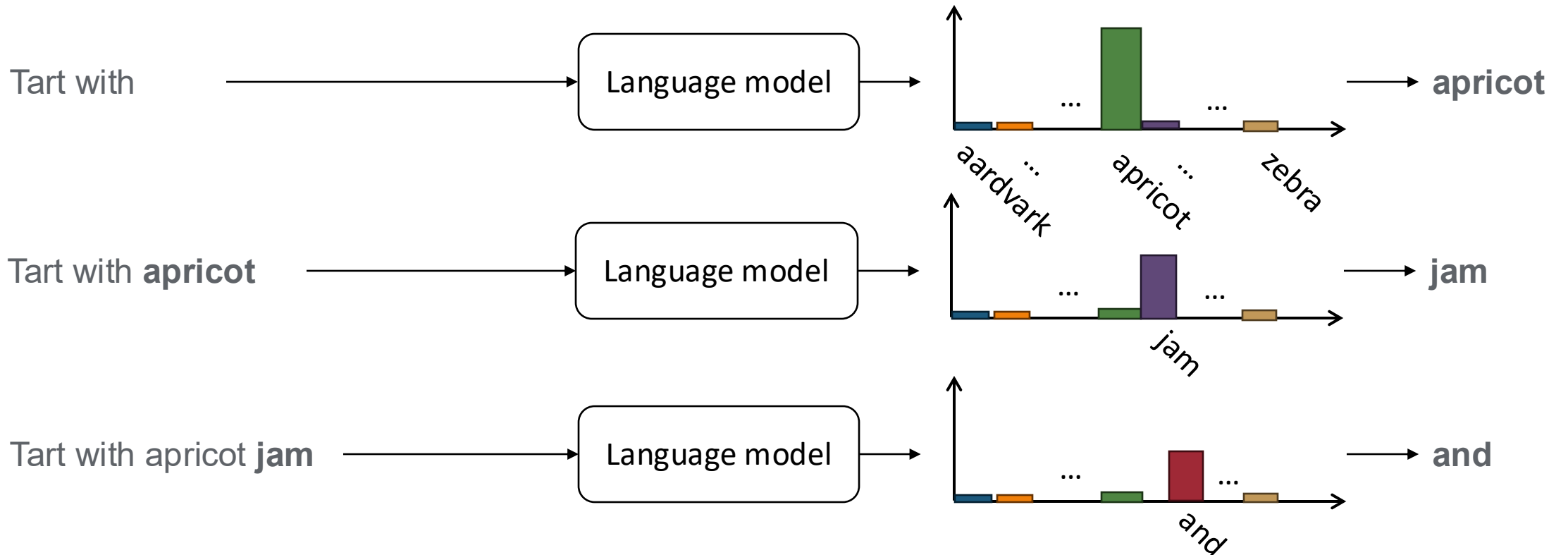
Two options:

Greedy approach: take as predicted word the highest probable

Sampling: sample (chose randomly) from the probability distribution

Then:

Add the predicted word to the input and continue predicting the next one



A few notes....

Large Language Models use efficient training techniques to elaborate many windows in parallel

Large Language Models sometimes divide words in pieces (***tokens***). For simplicity we will consider tokens as just words

We can give to a language model a huge amount of text as training data: ***self-supervised learning***

Let's open the box: Transformers

BERT-like

<https://github.com/google-research/bert>
(Auto-encoding transformer models)

GPT-like

<https://github.com/openai/gpt-2>
(Auto-regressive transformer models)

BART-like

<https://github.com/pytorch/fairseq/tree/main/examples/bart>
(Seq2Seq transformer models)

<https://github.com/huggingface/transformers>

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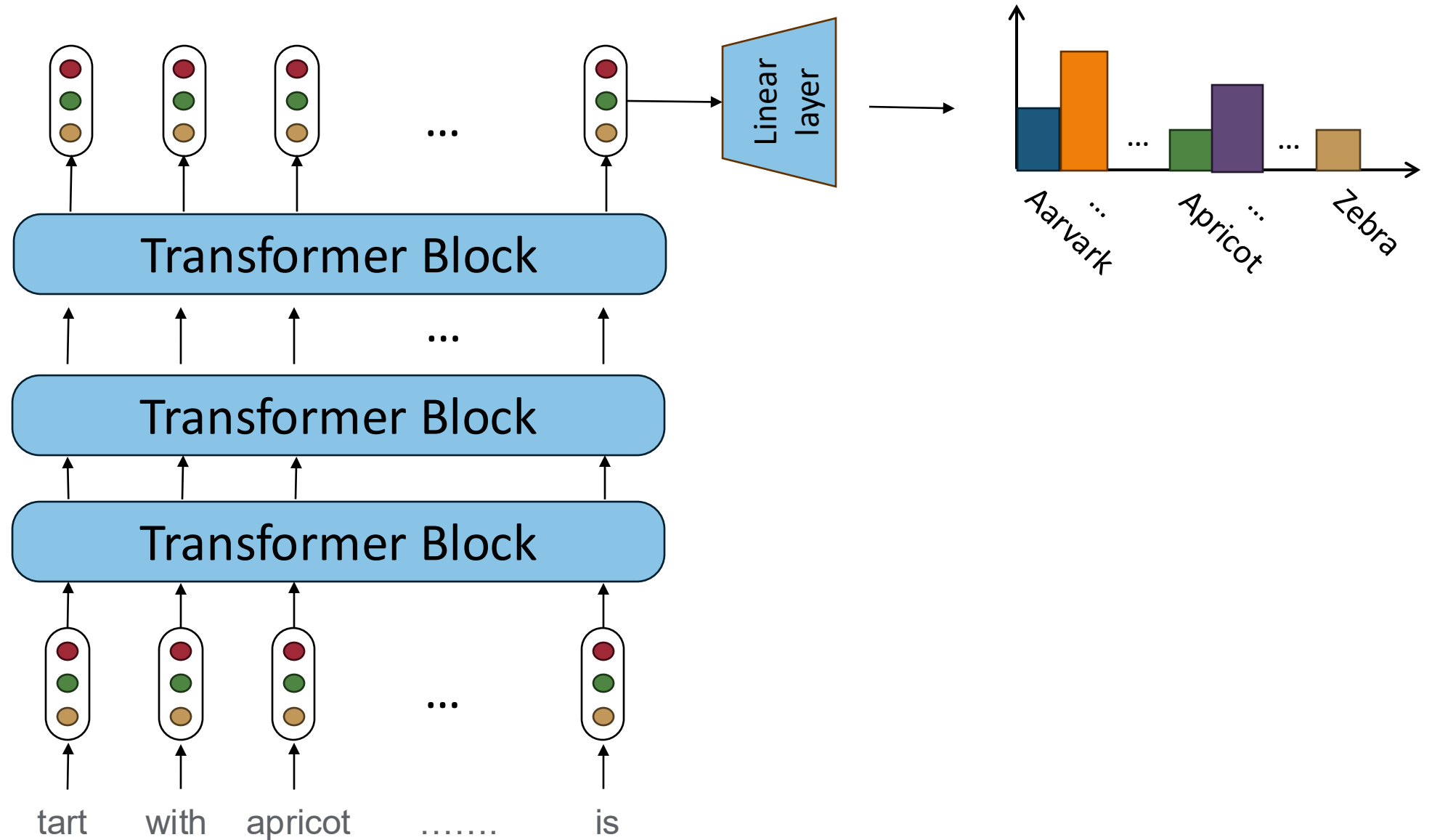
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(Auto-regressive transformer models)

BART-like

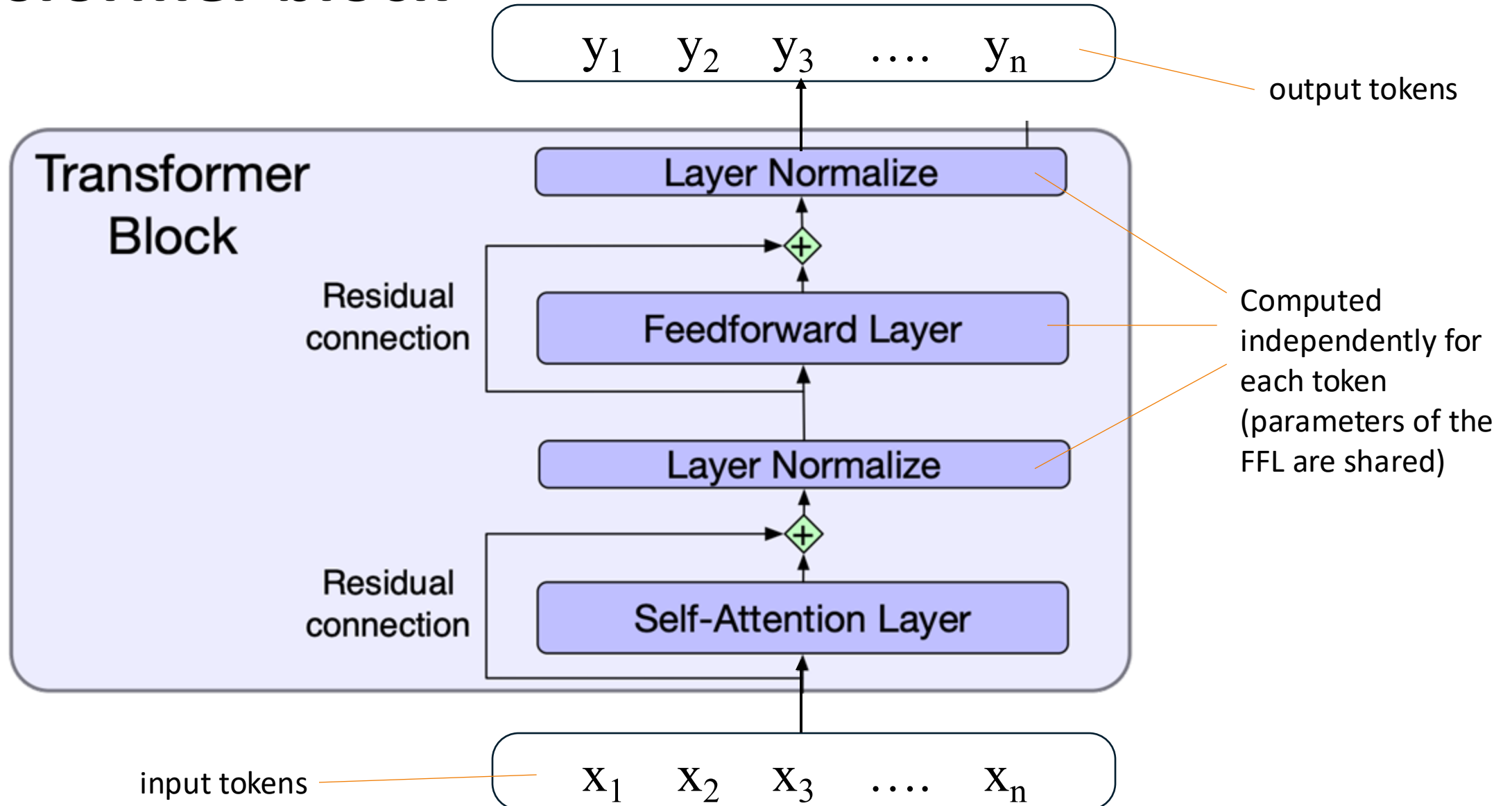
<https://github.com/pytorch/fairseq/tree/main/examples/bart>
(Seq2Seq transformer models)

<https://github.com/huggingface/transformers>

Transformer to predict the next word

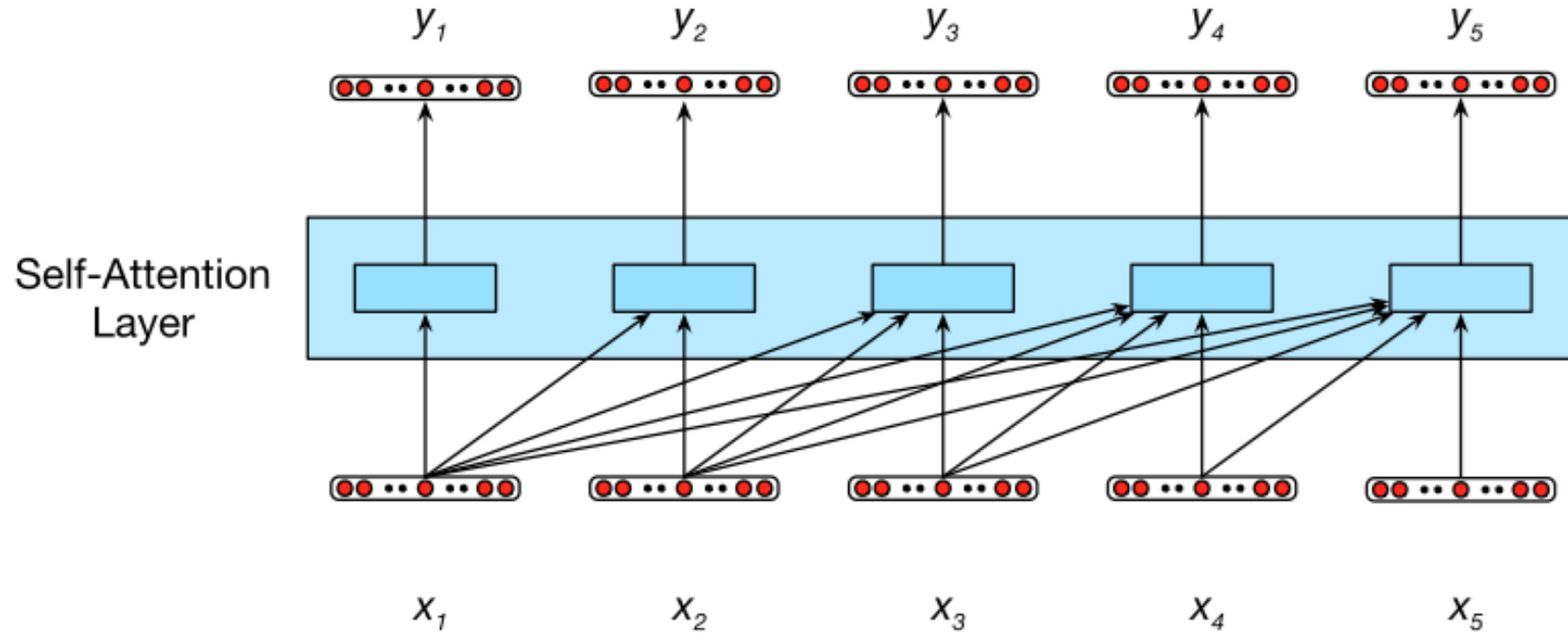


Transformer block

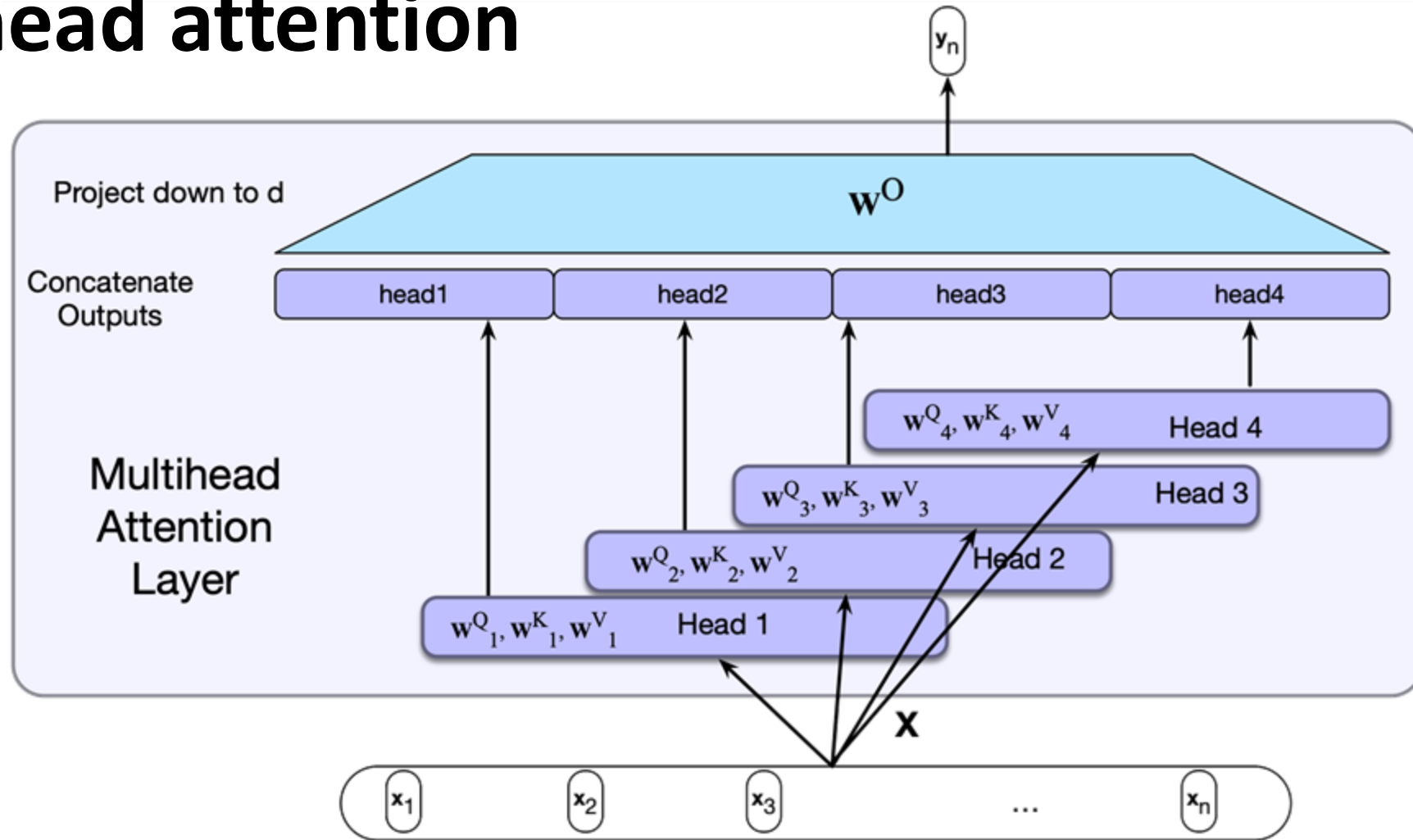


Self-attention layer

Transformers are based on the attention mechanism. The main component is the **self-attention** layer



Multi-head attention



The output of each head is of size d/h



Training Large Language Models: Models: The GPT-3 Example

Dataset Scale

GPT-3 was trained on a 300 billion
billion tokens training dataset

Dataset	Quantity (tokens)	Weight in training mix	Epochs elapsed when training for 300B tokens
Common Crawl (filtered)	410 billion	60%	0.44
WebText2	19 billion	22%	2.9
Books1	12 billion	8%	1.9
Books2	55 billion	8%	0.43
Wikipedia	3 billion	3%	3.4



Key Training Datasets for Modern LLMs

WebText Dataset

Curated from Reddit links with 3+ karma scores, yielding 8 million high-quality quality documents totaling 40 GB of text.

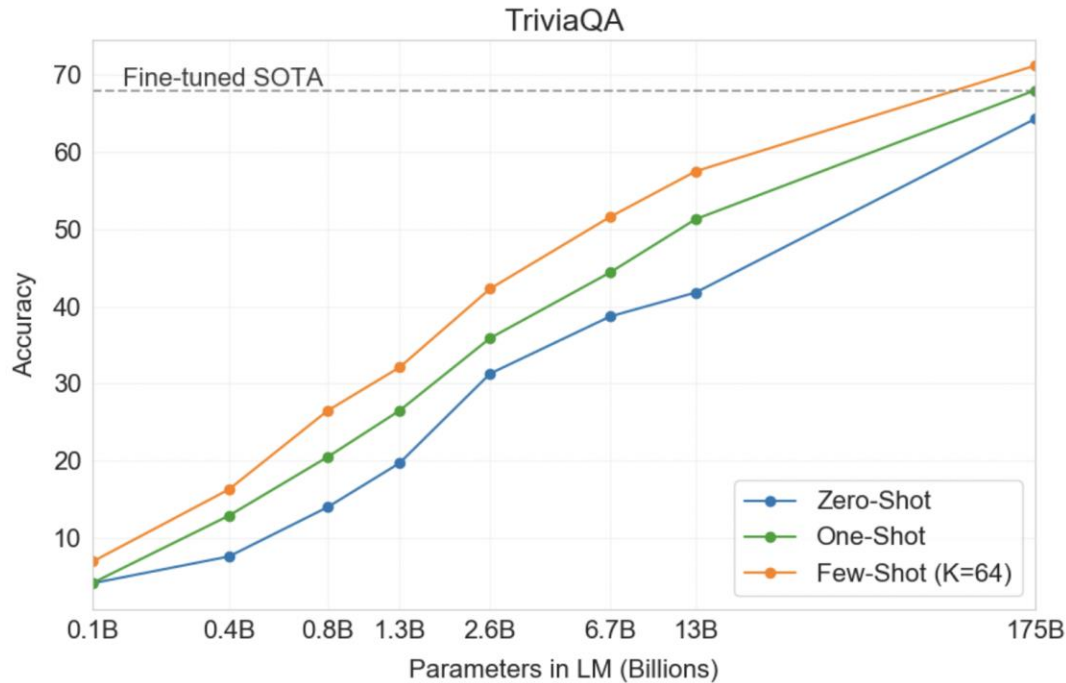
- Wikipedia excluded to prevent train-test overlap
- Byte Pair Encoding tokenization with modifications

CommonCrawl Repository

Open web crawl data spanning 16 years with over 250 billion pages and monthly monthly additions of 3-5 billion new pages.

- Quality filtering using WebText classifier
- Deduplication to remove overlapping content

Performance vs. model size

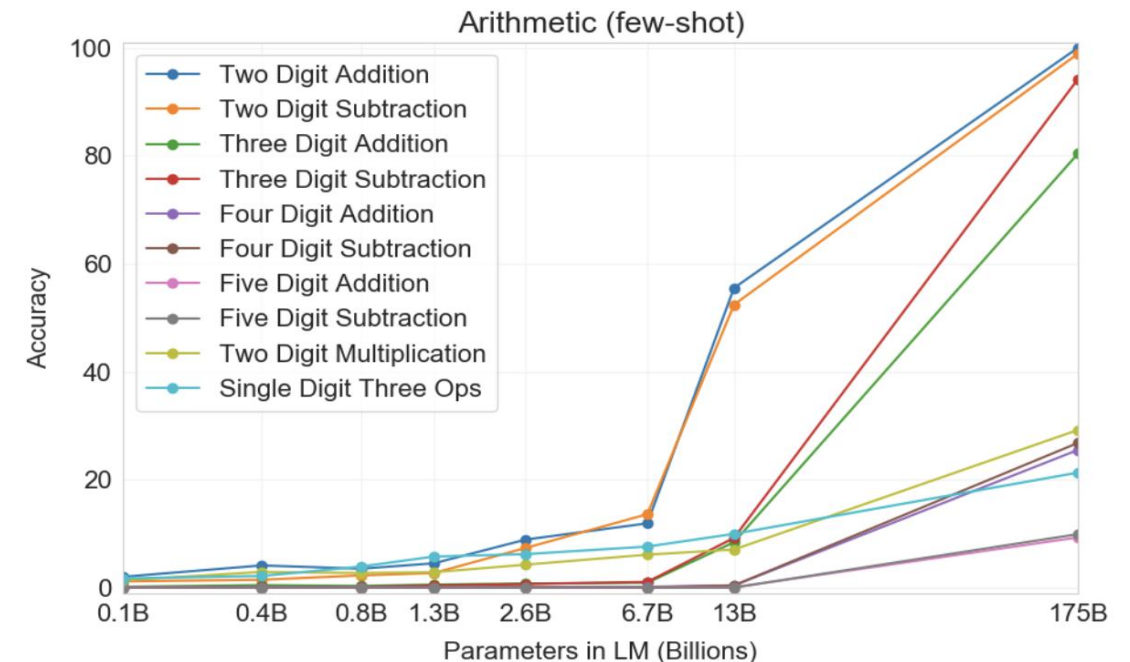


Emergent abilities: by increasing the model size, once a threshold is crossed, new abilities emerge

E.g. arithmetic abilities

Scalability: the ability to solve tasks without specific training increases with the number of model parameters.

Cognitive abilities scale with appropriate computing resources.

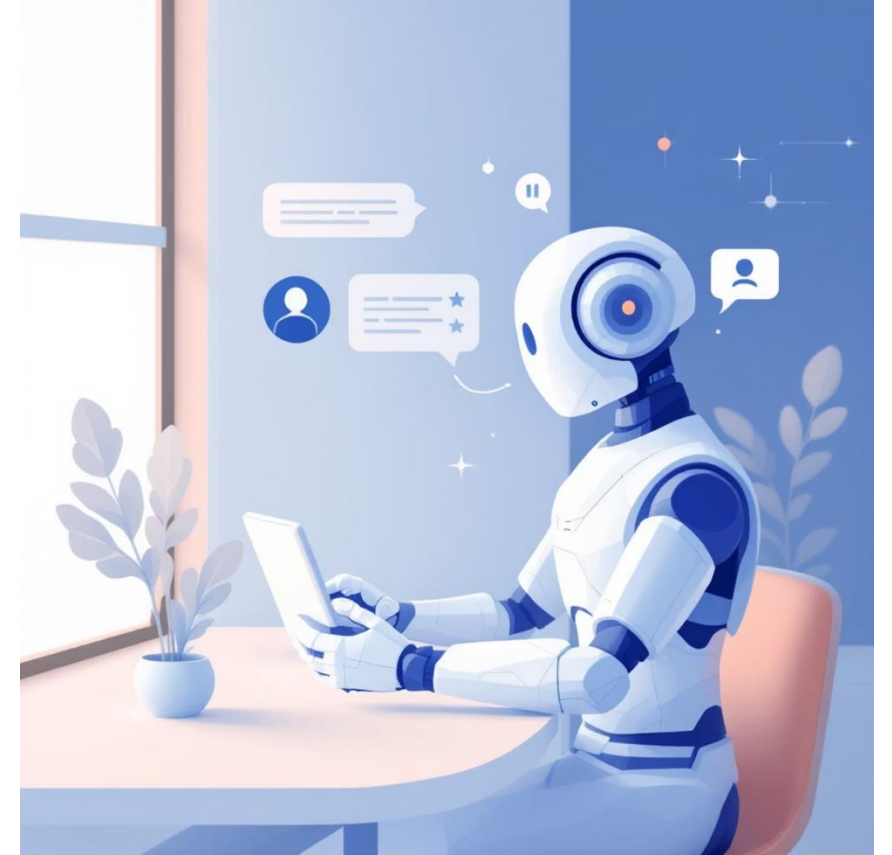


How to make an LLM follow instructions

So far...

Given partial text, LLMs can generate meaningful continuations. When presented with questions, they produce likely answers based on training patterns.

❓ **Critical Question:** How can we teach LLMs to follow specific instructions rather than just predict likely text?



InstructGPT

"Training language models to follow instructions with human feedback"

Long Ouyang, Jeff Wu, et al. | NeurIPS 2022

Training language models to follow instructions with human feedback

Long Ouyang* Jeff Wu* Xu Jiang* Diogo Almeida* Carroll L. Wainwright*
Pamela Mishkin* Chong Zhang Sandhini Agarwal Katarina Slama Alex Ray
John Schulman Jacob Hilton Fraser Kelton Luke Miller Maddie Simens
Amanda Askell[†] Peter Welinder Paul Christiano*[‡]
Jan Leike* Ryan Lowe*

OpenAI

Abstract

Making language models bigger does not inherently make them better at following a user's intent. For example, large language models can generate outputs that are untruthful, toxic, or simply not helpful to the user. In other words, these models are not *aligned* with their users. In this paper, we show an avenue for aligning language models with user intent on a wide range of tasks by fine-tuning with human feedback. Starting with a set of labeler-written prompts and prompts submitted through the OpenAI API, we collect a dataset of labeler demonstrations of the desired model behavior, which we use to fine-tune GPT-3 using supervised learning. We then collect a dataset of rankings of model outputs, which we use to further fine-tune this supervised model using reinforcement learning from human feedback. We call the resulting models *InstructGPT*. In human evaluations on our prompt distribution, outputs from the 1.3B parameter InstructGPT model are preferred to outputs from the 175B GPT-3, despite having 100x fewer parameters. Moreover, InstructGPT models show improvements in truthfulness and reductions in toxic output generation while having minimal performance regressions on public NLP datasets. Even though InstructGPT still makes simple mistakes, our results show that fine-tuning with human feedback is a promising direction for aligning language models with human intent.

✓ **Impact:** This foundational research directly led to the development of ChatGPT and modern instruction-following AI systems.



The Alignment Problem

Undesired Behaviors

Large language models frequently generate toxic, untruthful, or harmful content that directly contradicts user intentions and societal values

The Alignment Challenge

Models are not **aligned** with human values and user expectations, creating a critical gap between capability and safety

Solution Requirement

We need systematic methods to align model behavior with human feedback, ensuring responsible and helpful AI interactions

InstructGPT Training Methodology

InstructGPT builds on GPT-3 architecture with a sophisticated two-phase fine-tuning protocol:



Supervised Fine-Tuning (SFT)

1. Recruit and screen human annotators
2. Collect high-quality demonstration prompts and responses
3. Fine-tune base model on these examples



Reinforcement Learning from Human Feedback (RLHF)

4. Generate multiple responses using SFT model
5. Human annotators rank response quality
6. Train a Reward Model to automate quality assessment
7. Apply Proximal Policy Optimization (PPO) for final training



Key Insight: RLHF can be iterated multiple times for continuous improvement.

1. Annotator Selection Process

Human labelers are recruited through freelance platforms and undergo rigorous screening to ensure alignment with research objectives:

01

Sensitive Speech Flagging

Ability to consistently identify inappropriate content across various contexts and cultural sensitivities.

02

Response Ranking Agreement

Consensus with researchers on quality and preference judgments for model outputs.

03

Demonstration Writing

Skill in crafting high-quality prompt-response pairs that exemplify desired model behavior.

04

Diverse Perspective Assessment

Self-evaluated ability to recognize sensitivity across different demographic groups and communities.



2. Human-Written Training Examples

Annotators create demonstration prompt-response pairs to initialize the supervised learning process with high-quality examples:

"Labeler-written prompt and demonstration"

These carefully crafted examples serve as the foundation for teaching the model model appropriate response patterns and behavioral norms.

Prompt:

Create a shopping list from this recipe:

Trim the ends off zucchini. Cut zucchini in half lengthwise; scoop out pulp, leaving 1/2-in. shells. Finely chop pulp. In a skillet, cook beef, zucchini pulp, onion, mushrooms and peppers over medium heat until meat is no longer pink; drain. Remove from the heat. Add 1/2 cup cheese, ketchup, salt and pepper; mix well. Spoon into the zucchini shells. Place in a greased 13x9-in. baking dish. Sprinkle with remaining cheese.

Labeler demonstration

ucchini, beef, onion, mushroom, peppers, cheese, ketchup, salt, pepper

3. Supervised Fine-Tuning Phase

Training Protocol

A GPT-3 model undergoes standard supervised fine-tuning using the curated prompt-response pairs from human annotators.

The resulting **SFT model** serves as both the baseline for comparison and the foundation for subsequent reinforcement learning phases.



4., 5. Human Response Ranking System

Submit Skip Page 3 / 11 Total time: 05:39

Instruction

Summarize the following news article:

====
{article}
====

Output A

summary1

Rating (1 = worst, 7 = best)

1 2 3 4 5 6 7

Fails to follow the correct instruction / task ? ☐ Yes ☐ No

Inappropriate for customer assistant ? ☐ Yes ☐ No

Contains sexual content ☐ Yes ☐ No

Contains violent content ☐ Yes ☐ No

Encourages or fails to discourage violence/abuse/terrorism/self-harm ☐ Yes ☐ No

Denigrates a protected class ☐ Yes ☐ No

Gives harmful advice ? ☐ Yes ☐ No

Expresses moral judgment ☐ Yes ☐ No

Notes

{Optional} notes

(a)

Ranking outputs

To be ranked

Rank 1 (best)	Rank 2	Rank 3	Rank 4	Rank 5 (worst)
1 A team of researchers from Yale University and University of California, Davis studied the vocalization patterns of several different types of parrots. They found that parrots like to mimic human speech, and can produce a wide range of sounds, such as whistles, squawks, and other types of vocalizations...	2 Parrots have been found to have the ability to understand numbers. Researchers have found that parrots can understand numbers up to six. In a series of experiments, the parrots were able to identify the amount of food items under a number of cups...	3 Scientists have found that green-winged parrots can tell the difference between two noises that are the same except for the order in which they are heard. This is important because green-winged parrots are known to imitate sounds. This research shows that they are able to understand the difference between sounds. 4 Current research suggests that parrots see and hear things in a different way than humans do. While humans see a rainbow of colors, parrots only see shades of red and green. Parrots can also see ultraviolet light, which is invisible to humans. Many birds have this ability to see ultraviolet light, an ability		

Evaluation Process

The SFT model generates multiple responses for each prompt, which human annotators then evaluate systematically.

Ranking Criteria

- **Usefulness:** How effectively the response addresses the prompt
- **Safety:** Absence of toxic or harmful content
- **Alignment:** Consistency with human values and expectations

Priority: Unharmful responses are consistently preferred over potentially useful but risky outputs.



6. Training Reward Model (RM)

A GPT-3 6B-based Reward Model is trained on human preference rankings to automate reward assessment. The RM takes a prompt-response pair as input and outputs a numerical reward value, enabling scalable evaluation of model outputs without constant human oversight.

This automated scoring system forms the foundation for reinforcement learning from human feedback (RLHF), transforming subjective human preferences into quantifiable training signals.

7. Reinforcement Learning with PPO



The objective function of PPO maximizes rewards from the RM while minimizing divergence from the original SFT model. This dual objective ensures the model improves human preference alignment without losing its foundational capabilities.

Performance

1.3B

InstructGPT Parameters

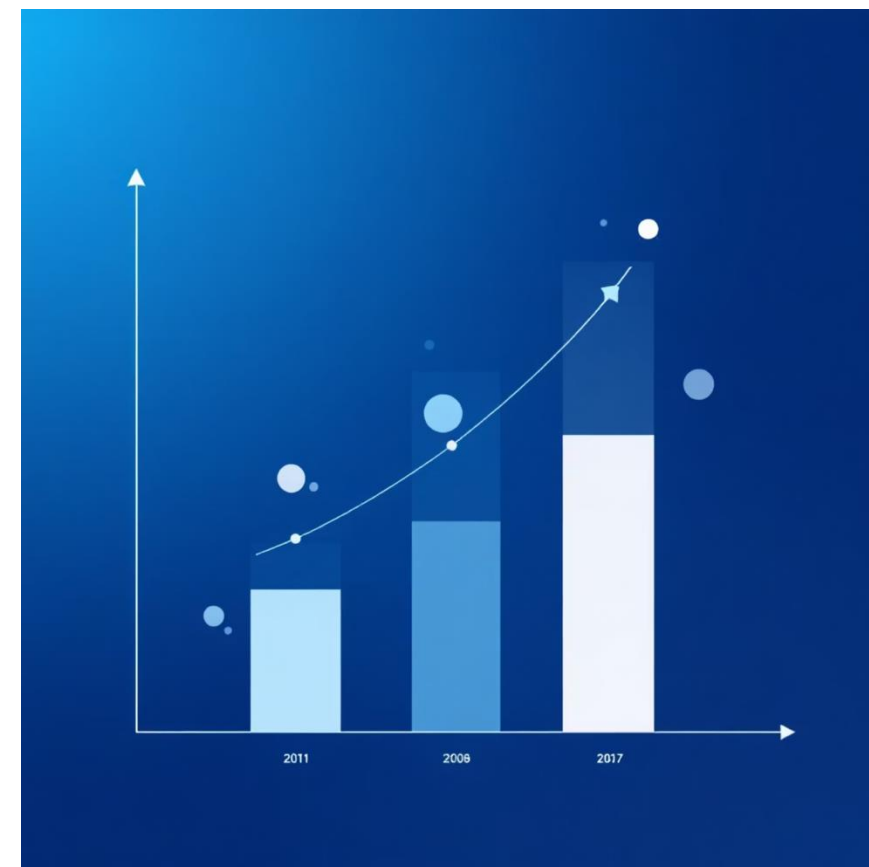
Preferred over 175B GPT-3

In human evaluations, outputs from the 1.3B parameter InstructGPT model are preferred to outputs from the 175B GPT-3, despite having 100x fewer parameters

100x

Fewer Parameters

Yet superior performance



InstructGPT models show significant improvements in truthfulness and dramatic reductions in toxic output generation, with minimal performance regressions on standard NLP benchmarks.

Using GPT Models in Practice

01

Explore the Playground

Try functionalities and get corresponding code at platform.openai.com/playground

02

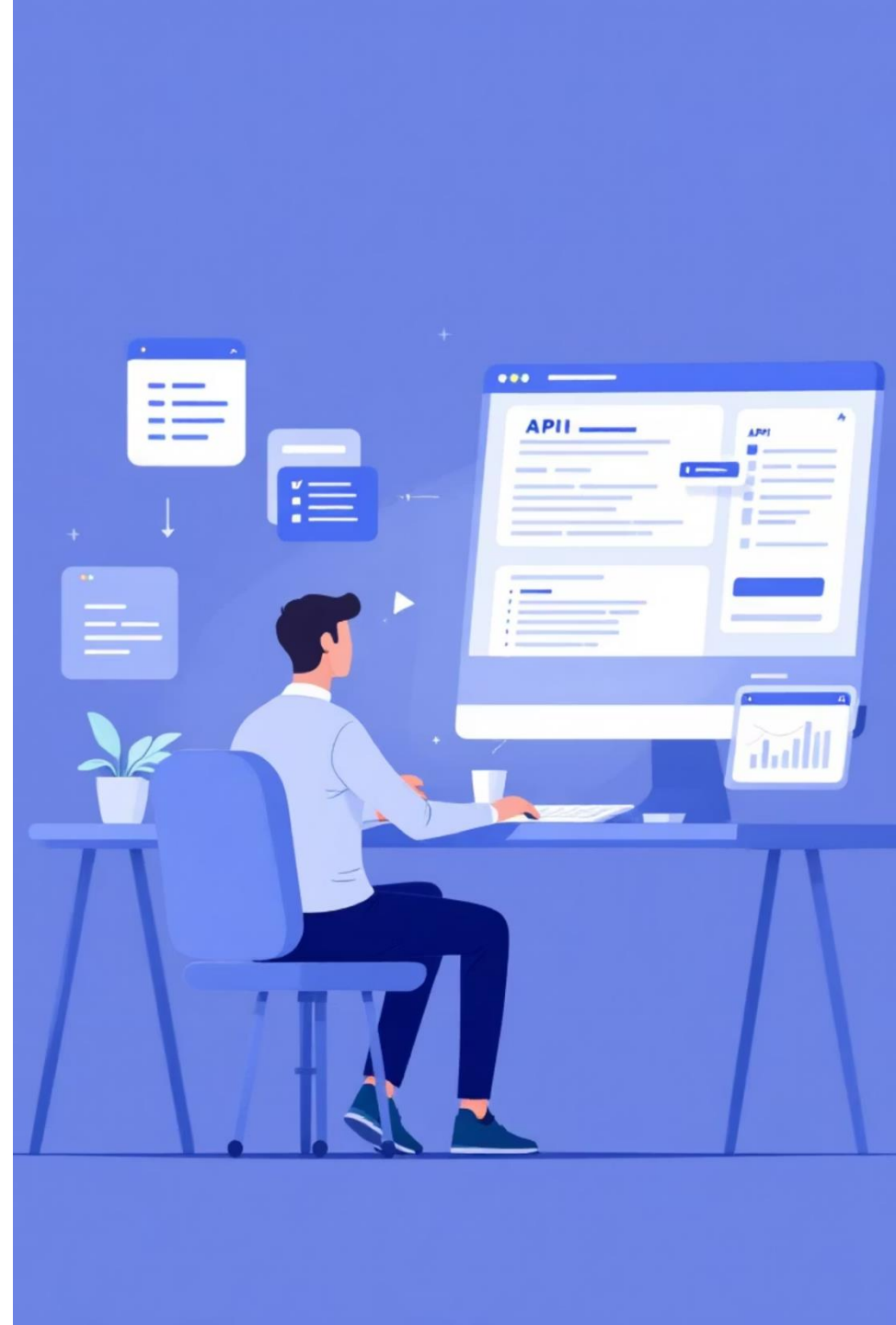
API Integration

Text generation guide: platform.openai.com/docs/guides/text-generation

03

Fine-tuning

Custom model training: platform.openai.com/docs/guides/fine-tuning





Open Source Alternative: Ollama

<https://ollama.com/>

Local Deployment

Run large language models locally without cloud dependencies or API costs

Privacy Control

Keep sensitive data on-premises while leveraging powerful language model capabilities

Model Variety

Access to multiple open-source models including Llama, Mistral, and CodeLlama CodeLlama variants

Ollama provides a freely available, self-hosted alternative to commercial LLM APIs, offering researchers and developers greater control over their AI infrastructure.



Controllable Large Language Models

Modern LLMs provide no guarantees on generated text structure or content compliance. However, many real-world applications require fine-grained control over outputs.



Syntactic Structure Enforcement

Ensuring generated text follows specific grammatical patterns or formatting rules

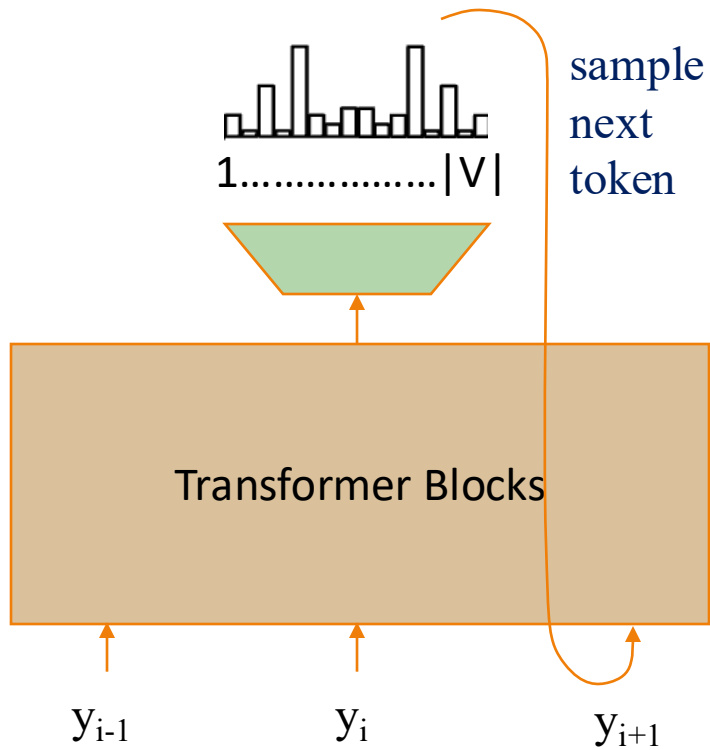


Taxonomy Adherence

Constraining outputs to predefined vocabularies or domain-specific specific terminologies

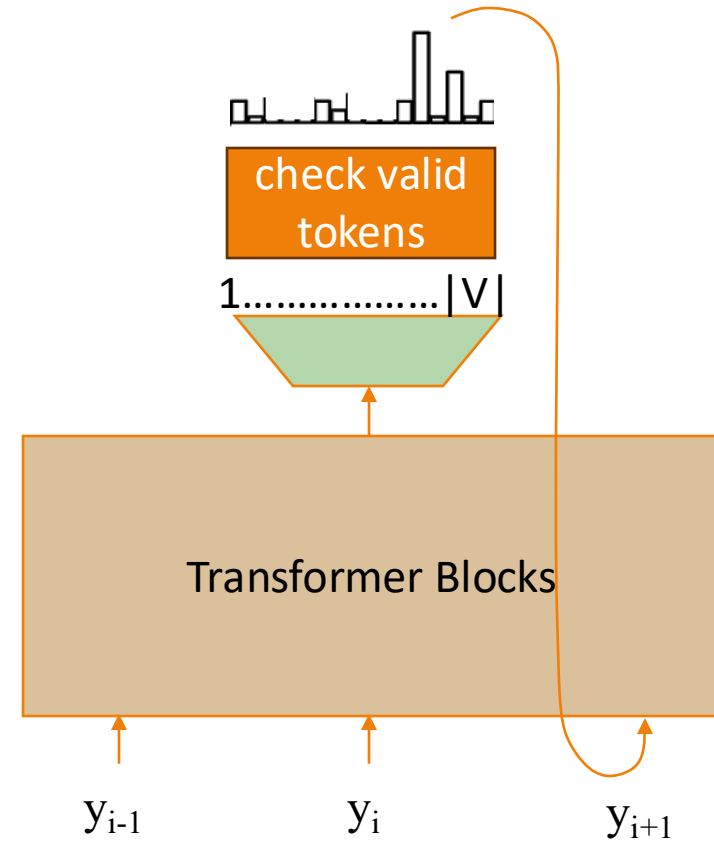
Standard vs. Constrained Decoding

Standard Decoding



Transformer blocks process tokens sequentially, sampling the next token from the full vocabulary without structural constraints.

Constrained Decoding



An additional validation step checks token validity before sampling, ensuring compliance with grammatical constraints throughout generation.



Token Validation with Automata



Finite State Automata (FSA)

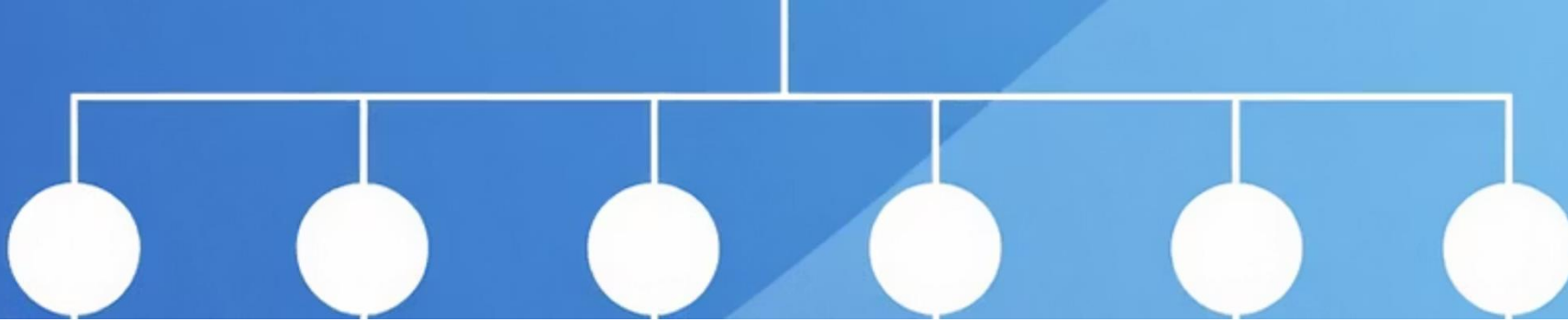
Recognizes regular languages with simple state transitions, suitable for basic pattern matching



Pushdown Automata (PDA)

More powerful than FSA, uses a stack to handle Context-Free Grammars (CFG) and nested structures

The validation process uses two key functions: `get_valid_tokens()` to filter allowable next tokens, and `update_state(yn)` to advance the automaton state after token selection



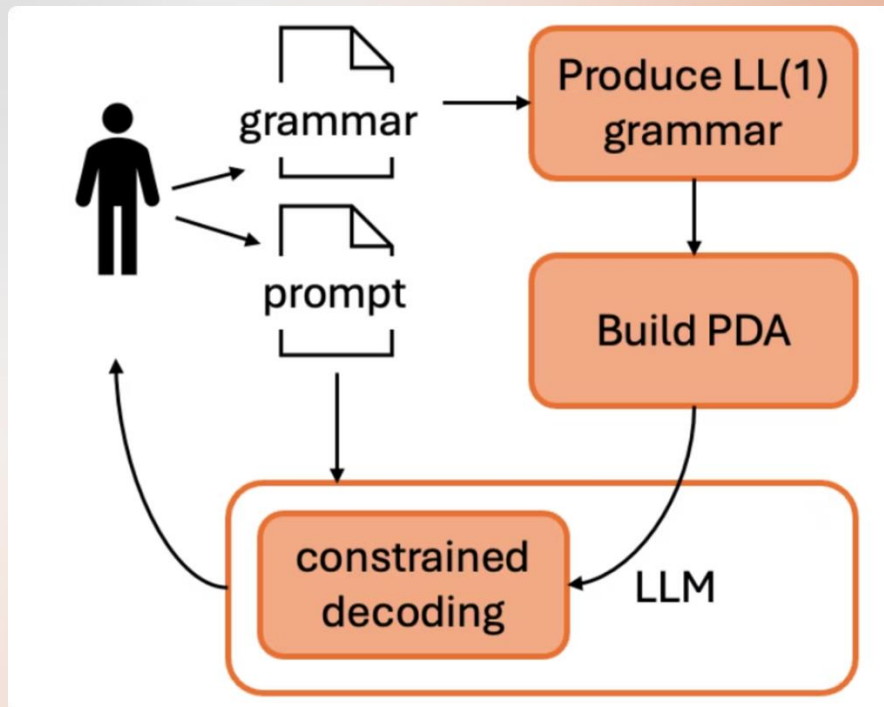
An Example of Context Free Grammar (CFG) for Hierarchical Classification

$S \rightarrow \text{biochemistry, } BiochemArea \mid$
 $\text{civil, } CivilArea \mid$
 $\text{cs, } CSArea \mid$
 $\text{ece, } ECEArea \mid$
 $\text{mae, } MAEArea \mid$
 $\text{medical, } MedicalArea \mid$
 $\text{psychology, } PsychologyArea$

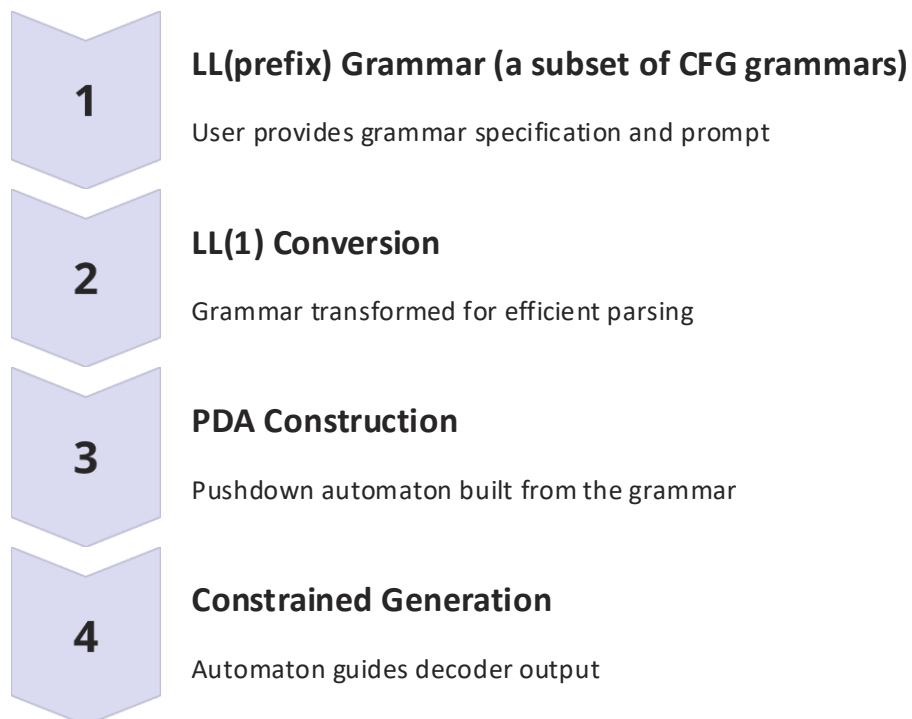
$BiochemArea \rightarrow \text{cell biology} \mid$
 $\text{dna/rna sequencing} \mid$
 $\text{enzymology} \mid$
 $\dots$
 $CivilArea \rightarrow \text{ambient intelligence} \mid$
 $\text{bamboo as a building material} \mid$
 $\text{construction management} \mid$
 $\dots$

Grammar rules define the hierarchical structure, ensuring generated classifications follow the specified taxonomy tree.

The resulting output maintains strict adherence to the classification hierarchy, preventing invalid preventing invalid category combinations.



GRAMMAR-LLM Architecture



The Grammar-LLM framework enables precise control over language model outputs by converting user-specified grammars into computational constraints that guide the generation process in real-time.

Tuccio, et al. "GRAMMAR-LLM: Grammar-constrained natural language generation" *Findings of the Association for Computational Linguistics: ACL 2025*. 2025.

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<https://platform.openai.com/docs/api-reference/chat/create>

Ollama <https://ollama.com/>